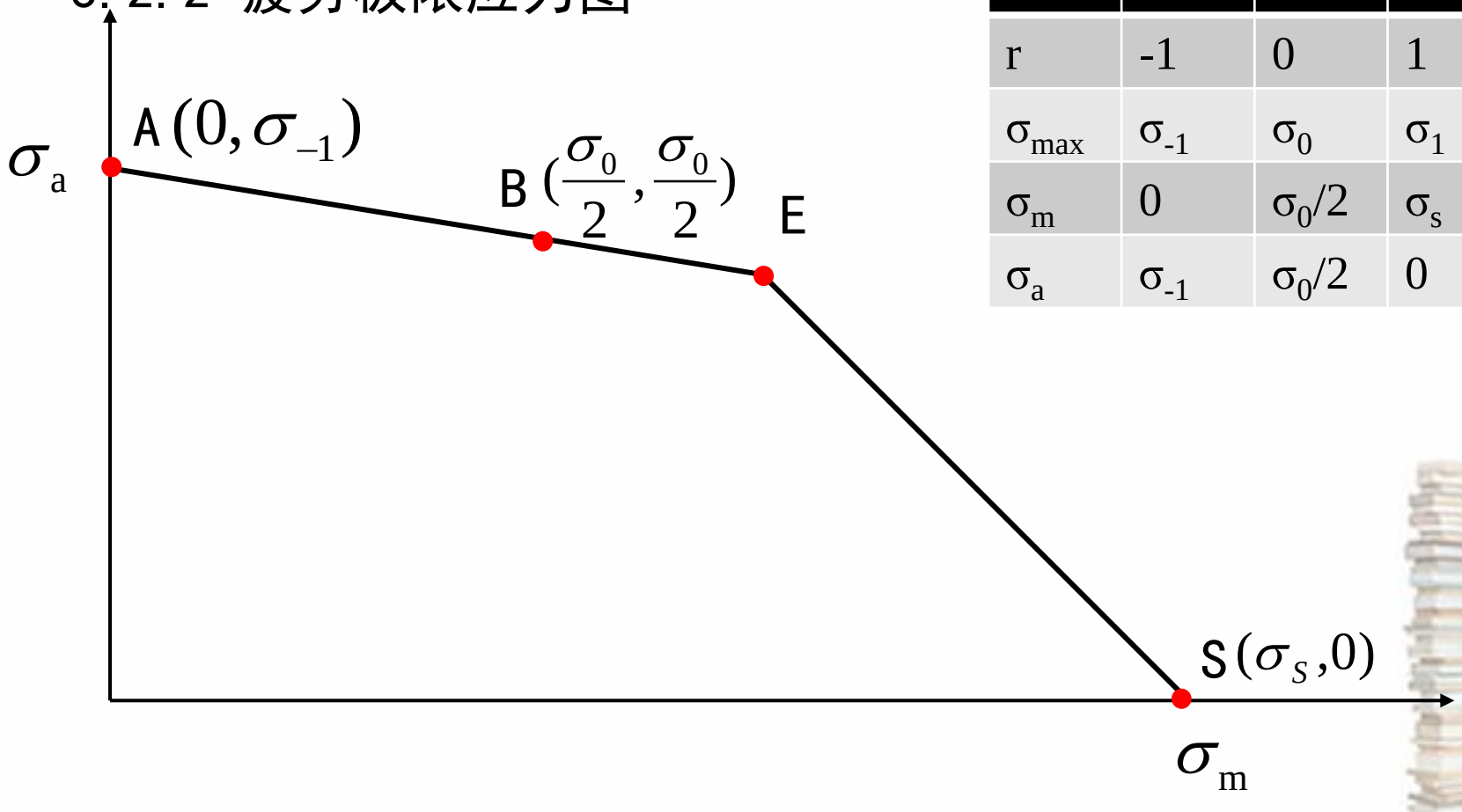
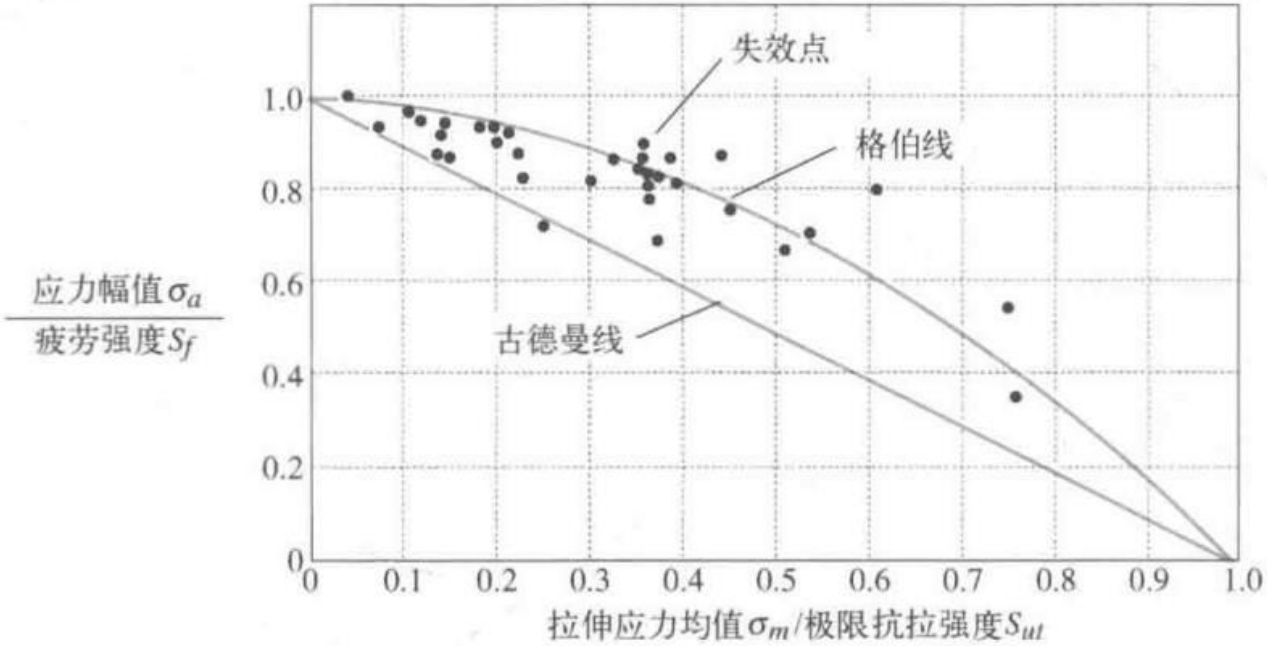
第3章 机械零件的疲劳强度

3.2 疲劳曲线和疲劳极限应力图

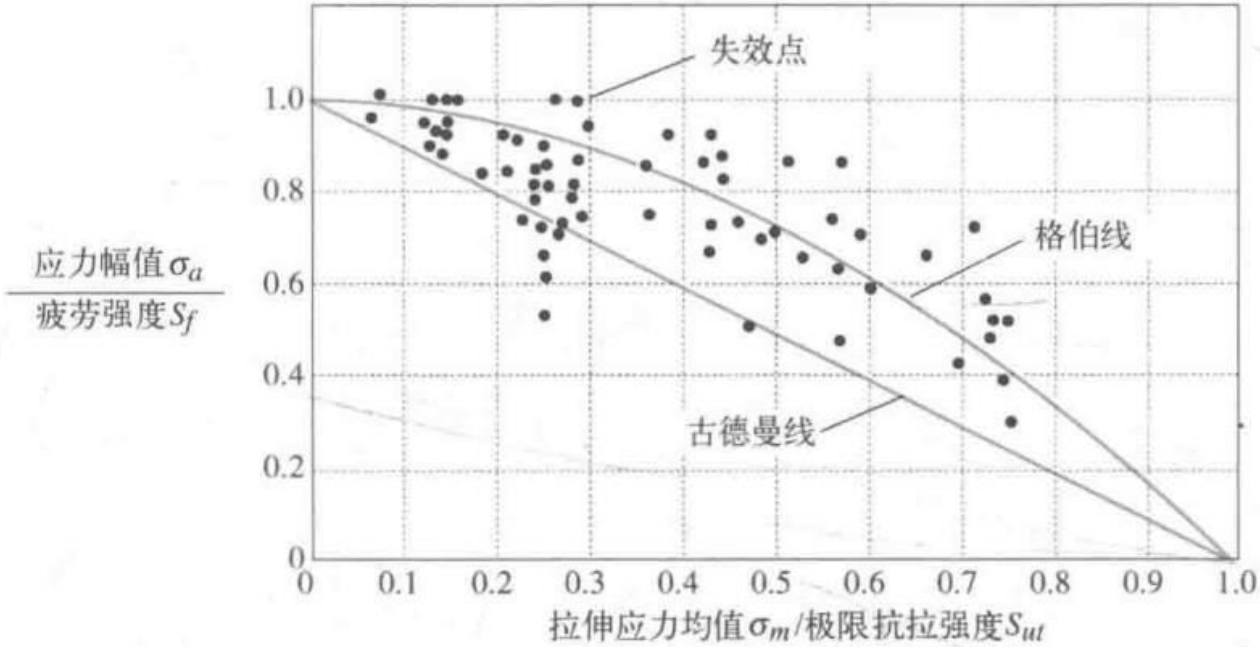
3.2.2 疲劳极限应力图



	A	B	S
r	-1	0	1
σ_{max}	σ_{-1}	σ_0	σ_1
σ_m	0	$\sigma_0/2$	σ_s
σ_a	σ_{-1}	$\sigma_0/2$	0



a) 钢，循环次数在10^7～10^8次



b) 铝合金，循环次数在5×10^8次

3.2 疲劳曲线和疲劳极限应力图

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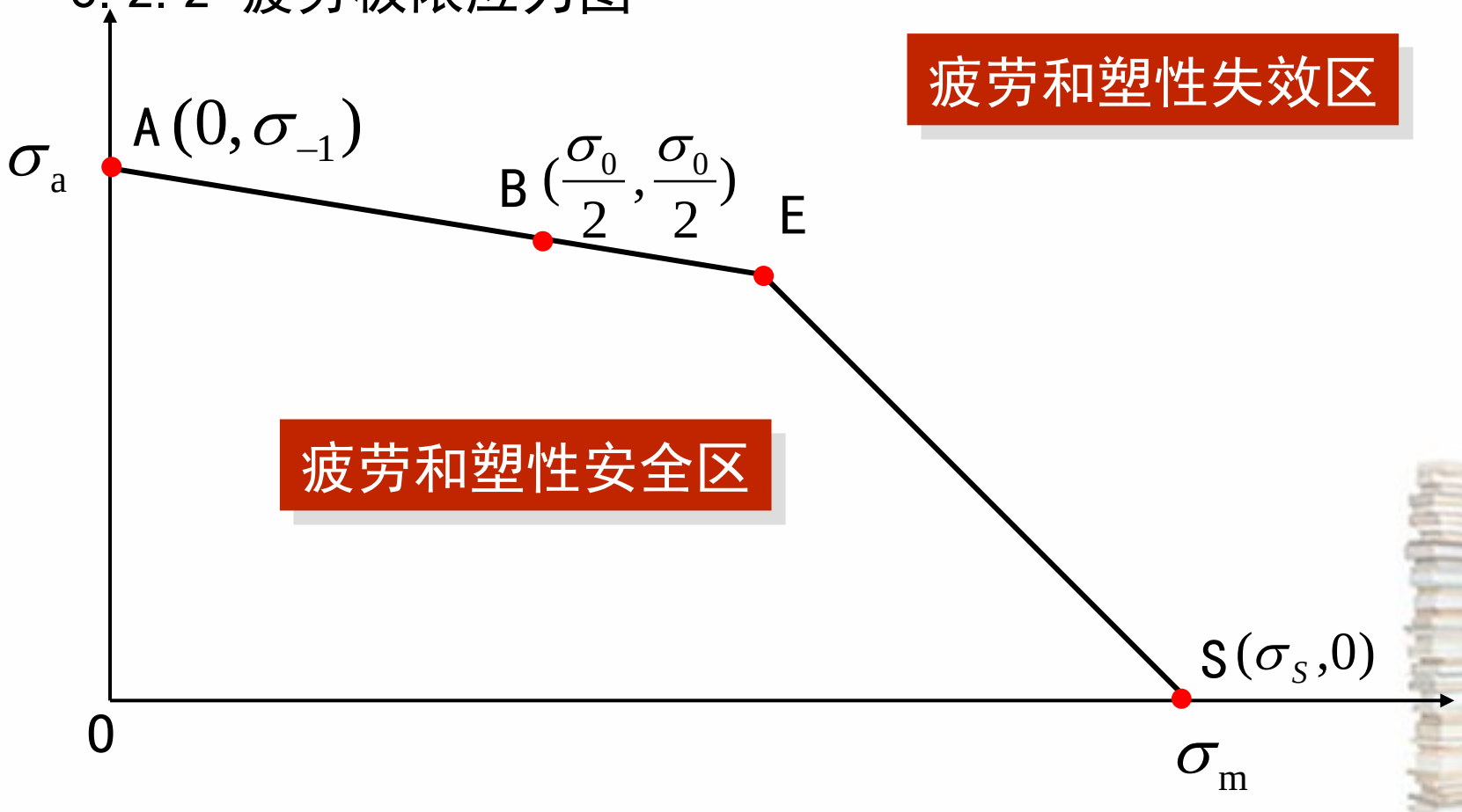

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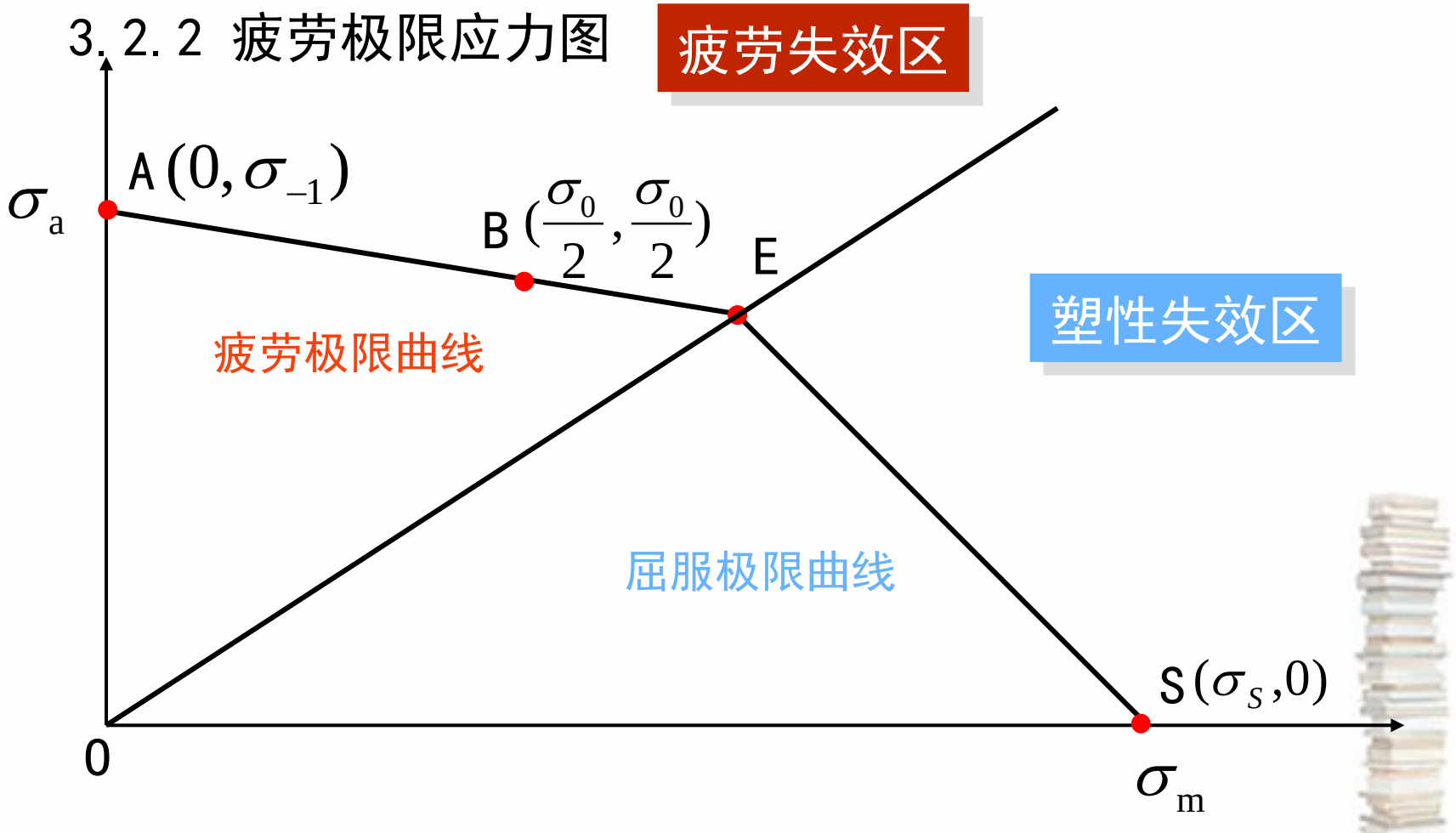


3.3 影响机械零件疲劳强度的主要因素

3.3.1 应力集中影响

$$K_\sigma = 1 + q(\alpha_\sigma - 1)$$

K_σ 越大，应力集中越严重，对疲劳强度的不良影响越显著

3.3 影响机械零件疲劳强度的主要因素

3.3.2 尺寸影响

ε_σ ε_τ

零件尺寸越大， ε_σ 越小，对疲劳强度的不良影响越显著

3.3 影响机械零件疲劳强度的主要因素

3.3.3 表面状态影响

β_σ β_τ

表面越粗糙， β_σ 越小，对疲劳强度的不良影响越显著

3.3 影响机械零件疲劳强度的主要因素

3.3.4 综合影响系数

$$(k_\sigma)_D = \frac{k_\sigma}{\varepsilon_\sigma \beta_\sigma}$$

对 σ_a 有影响

对 σ_m 没有影响

美国教材影响机械零件疲劳强度的主要因素，前三个相同，后面还有新增

温度系数　可靠度系数

$$S_e = C_{load} C_{size} C_{surf} C_{temp} C_{reliab} S_{e'}$$

$$S_f = C_{load} C_{size} C_{surf} C_{temp} C_{reliab} S_{f'}$$

载荷系数 C_{load} 为：

变矩：　$C_{load} = 1$

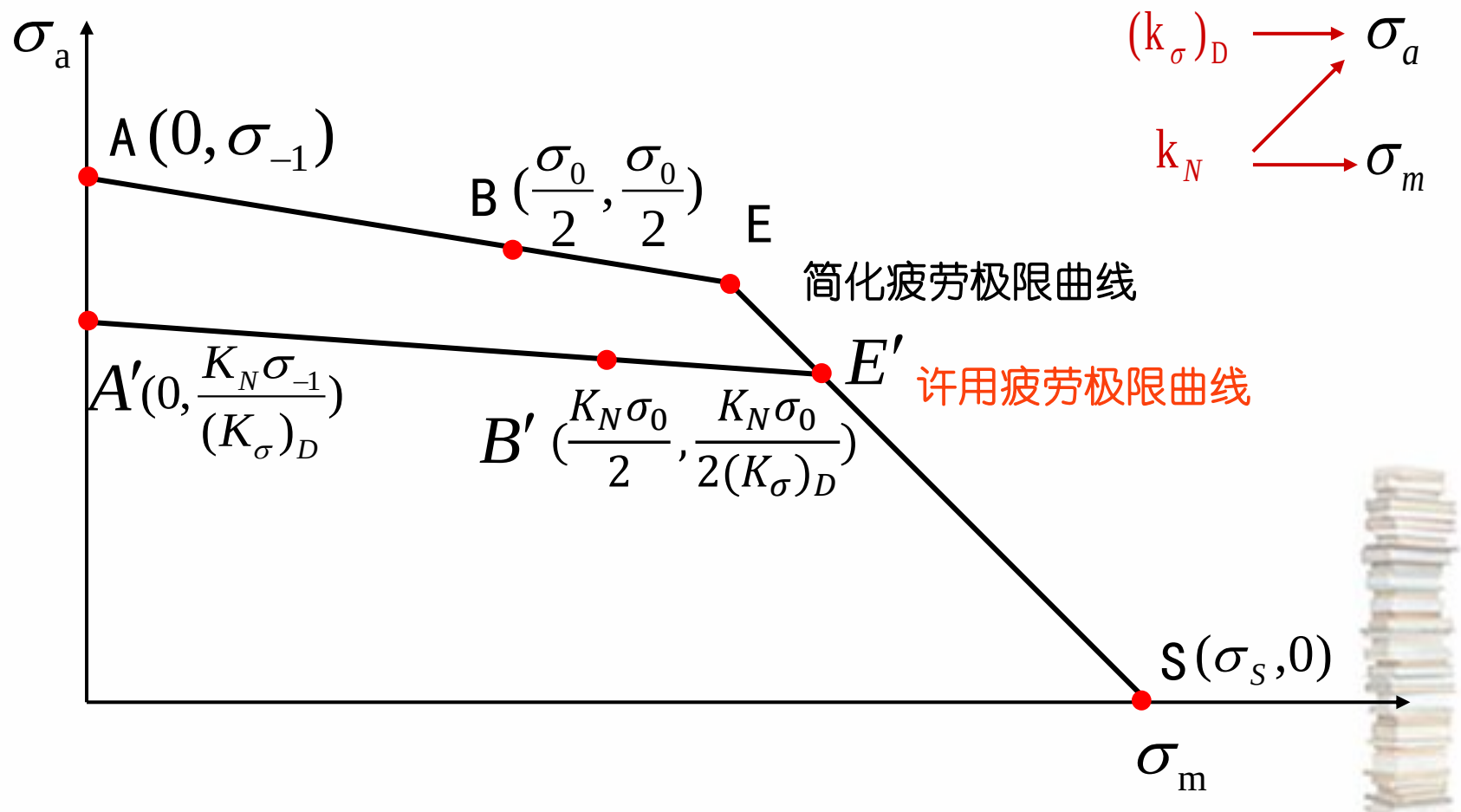
轴向载荷：　$C_{load} = 0.70$

当 $T \leqslant 450°C\ (840°F)$ 时，　$C_{temp} = 1$

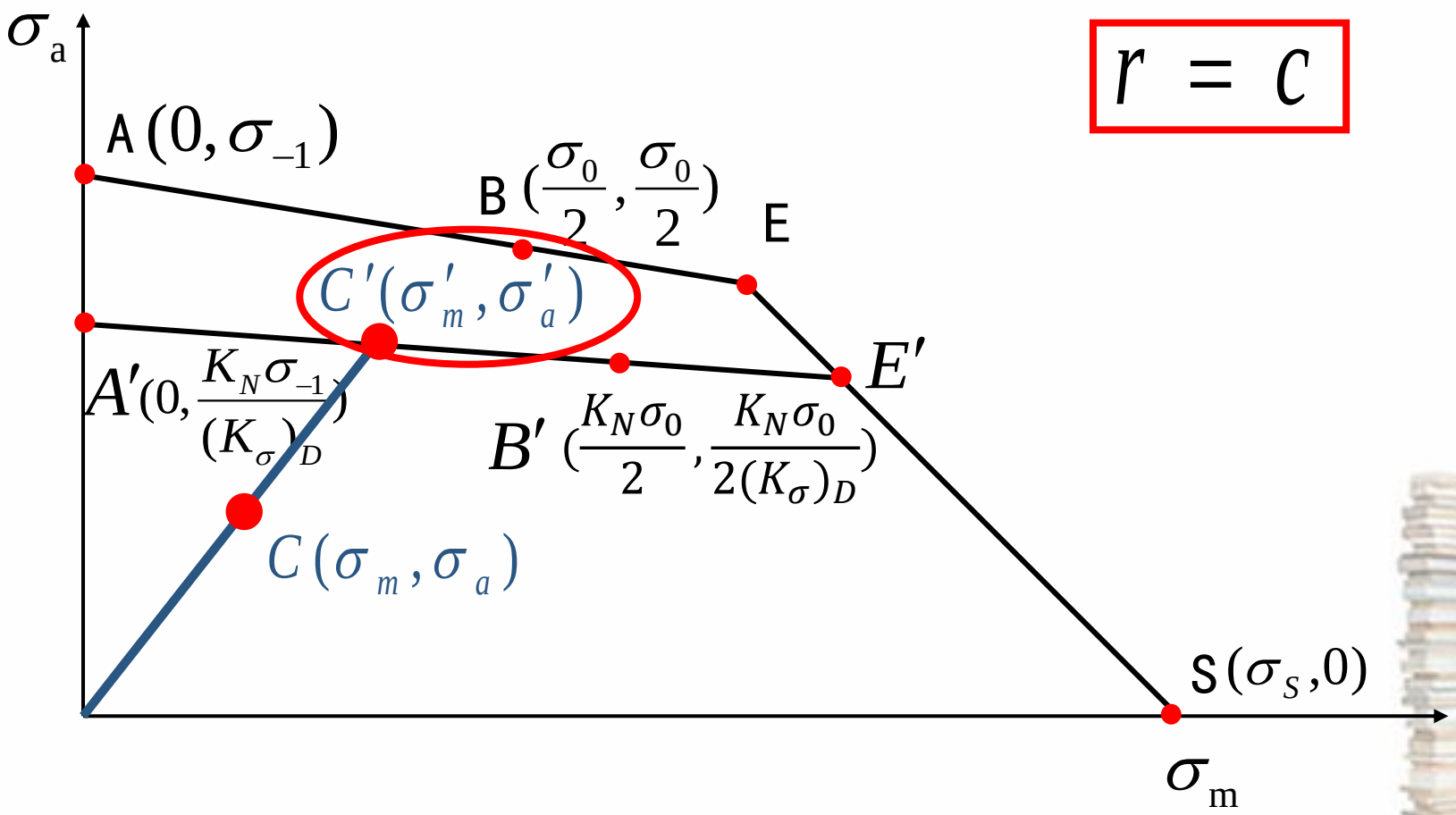
当 $450°C < T \leqslant 550°C$ 时，　$C_{temp} = 1 - 0.0058(T - 450)$

当 $840°F < T \leqslant 1\,020°F$ 时，　$C_{temp} = 1 - 0.0032(T - 840)$

3.4 许用疲劳极限应力图

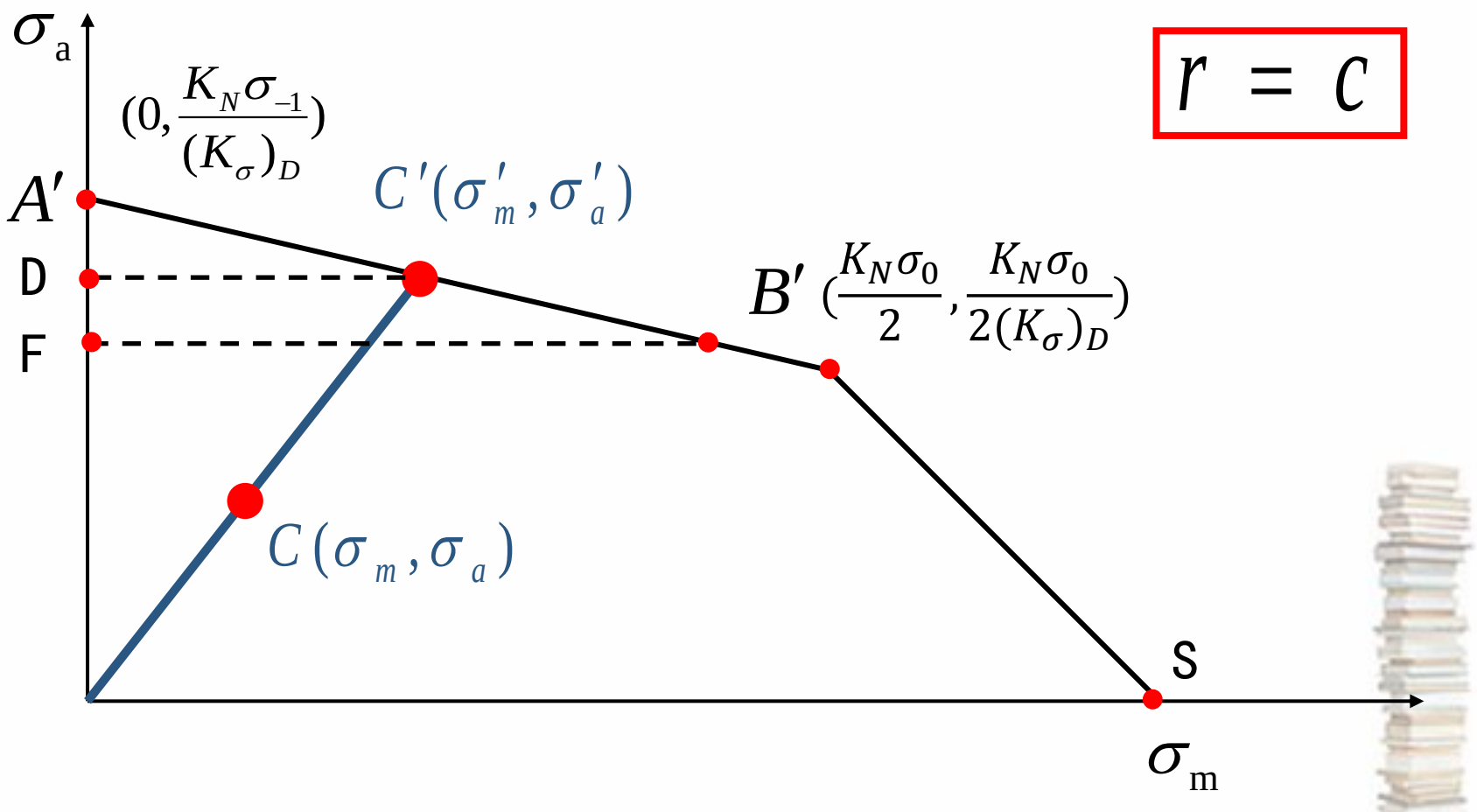


3.4 许用疲劳极限应力图



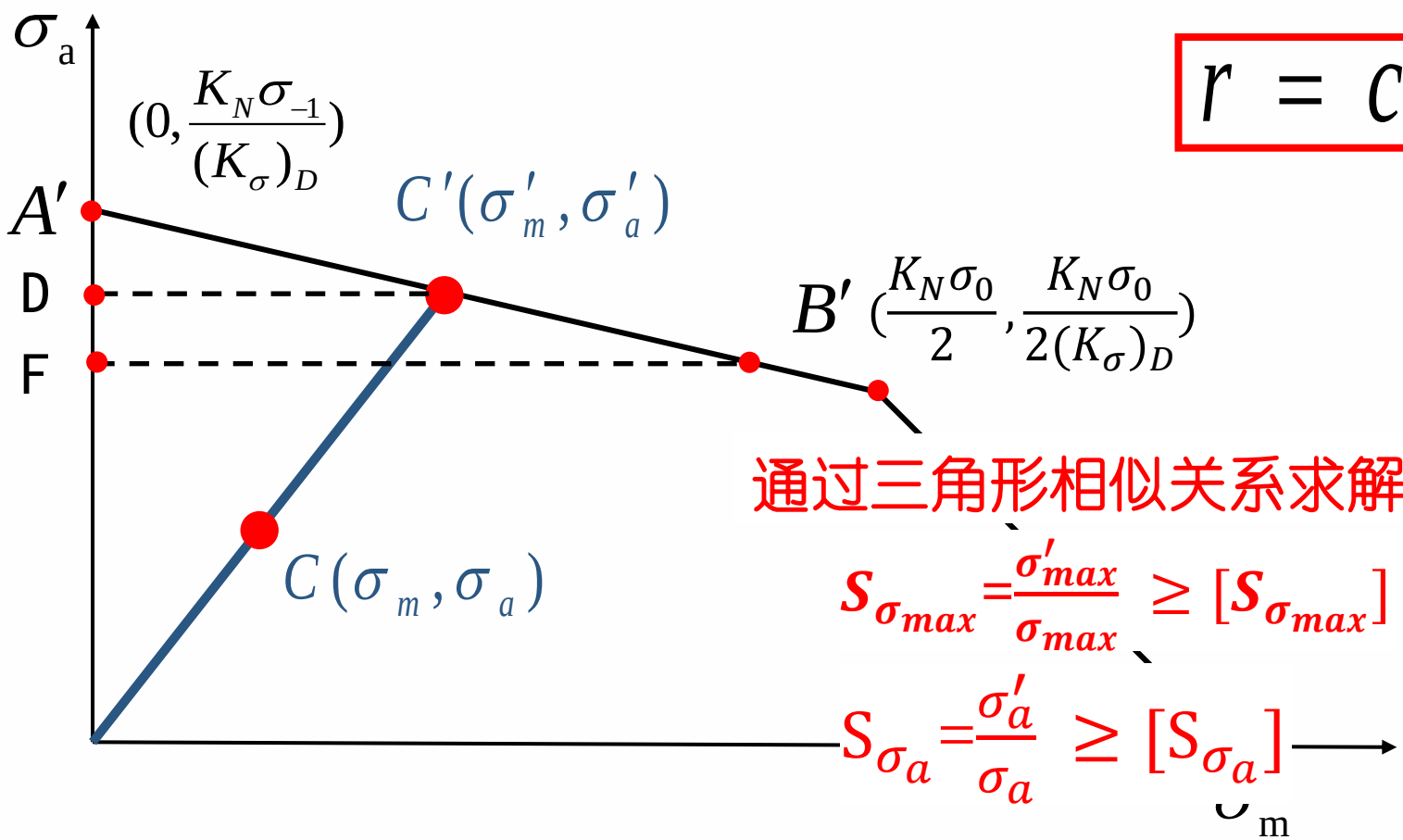
$$r = C$$

3.4 许用疲劳极限应力图



$$r = C$$

3.4 许用疲劳极限应力图



$$r = C$$

通过三角形相似关系求解安全系数

$$S_{\sigma_{max}}=\frac{\sigma'_{max}}{\sigma_{max}} \geq [S_{\sigma_{max}}]$$

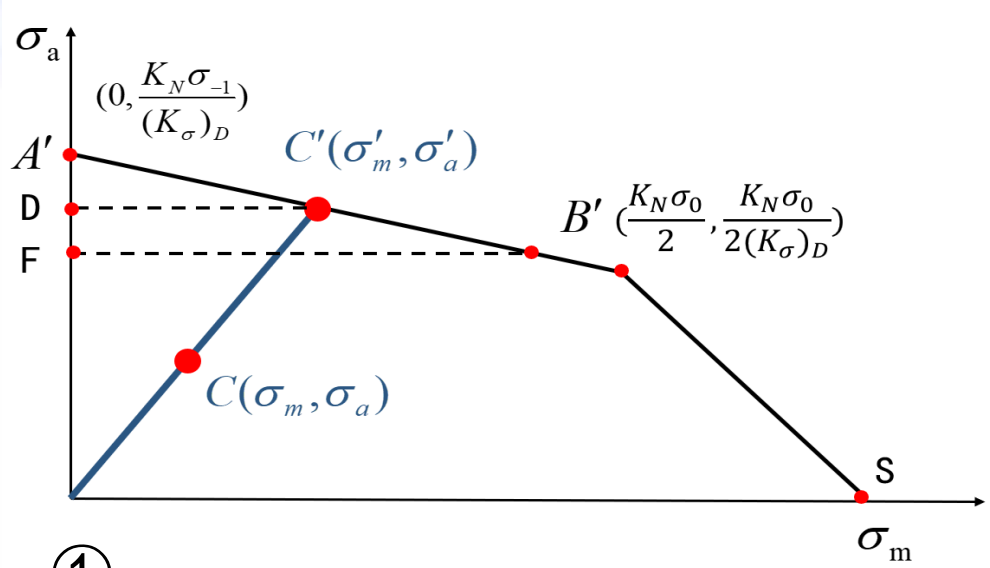
$$S_{\sigma_a}=\frac{\sigma'_a}{\sigma_a} \geq [S_{\sigma_a}]$$

3.4 许用疲劳极限应力图

$$\frac{A'D}{A'F}=\frac{DC'}{FB'}$$

$$\frac{\dfrac{K_N\sigma_{-1}}{(K_\sigma)_D}-\sigma'_a}{\dfrac{K_N\sigma_{-1}}{(K_\sigma)_D}-\dfrac{K_N\sigma_0}{2(K_\sigma)_D}}=\frac{\sigma'_m}{\dfrac{K_N\sigma_0}{2}}$$

$$\frac{\sigma_a}{\sigma'_a}=\frac{\sigma_m}{\sigma'_m}$$



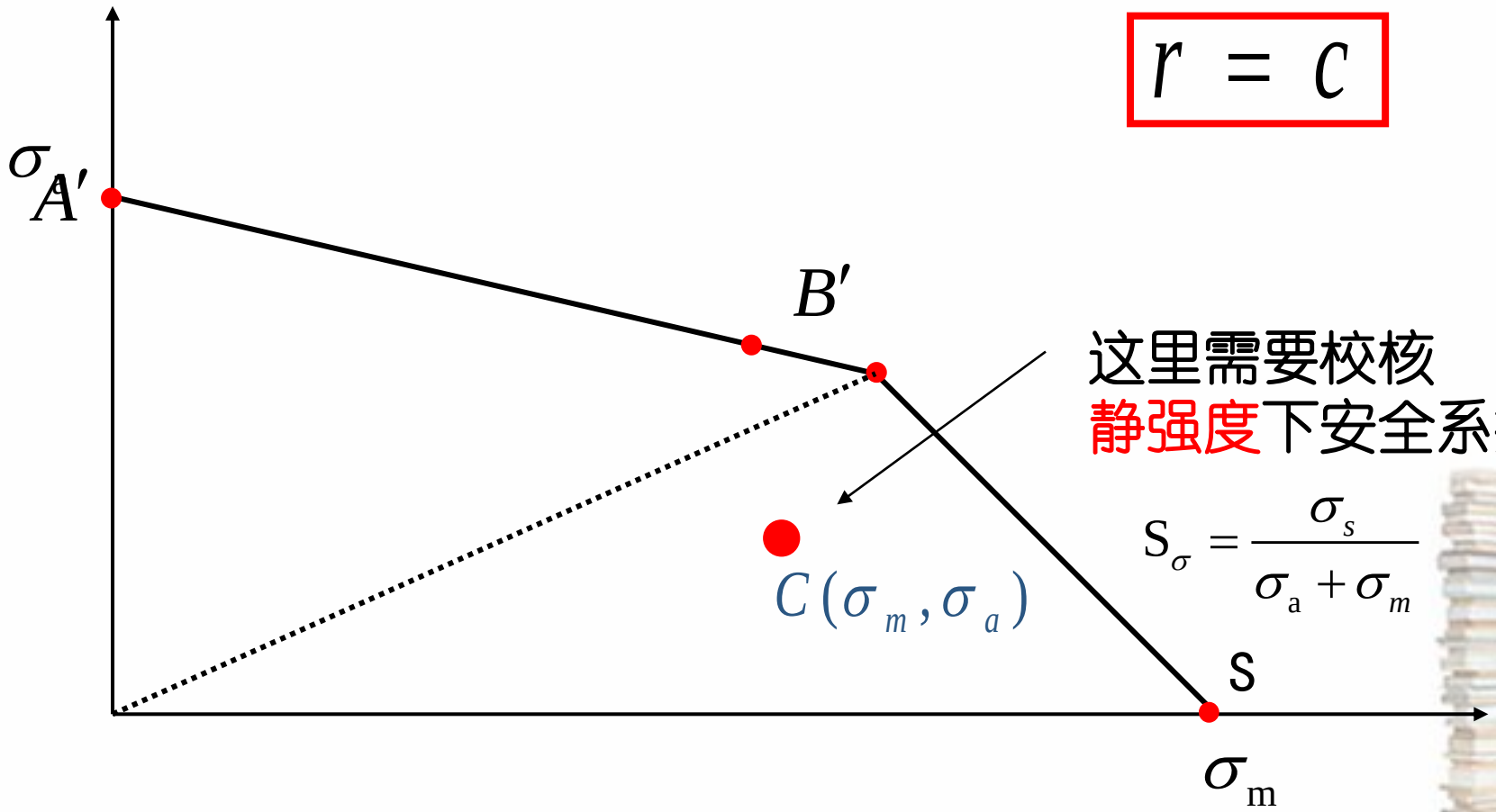
①

②

$$S_{\sigma_{max}}=S_{\sigma_a}=\frac{k_N\sigma_{-1}}{(k_\sigma)_D\sigma_a+\psi_\sigma\sigma_m}$$

3.4 许用疲劳极限应力图

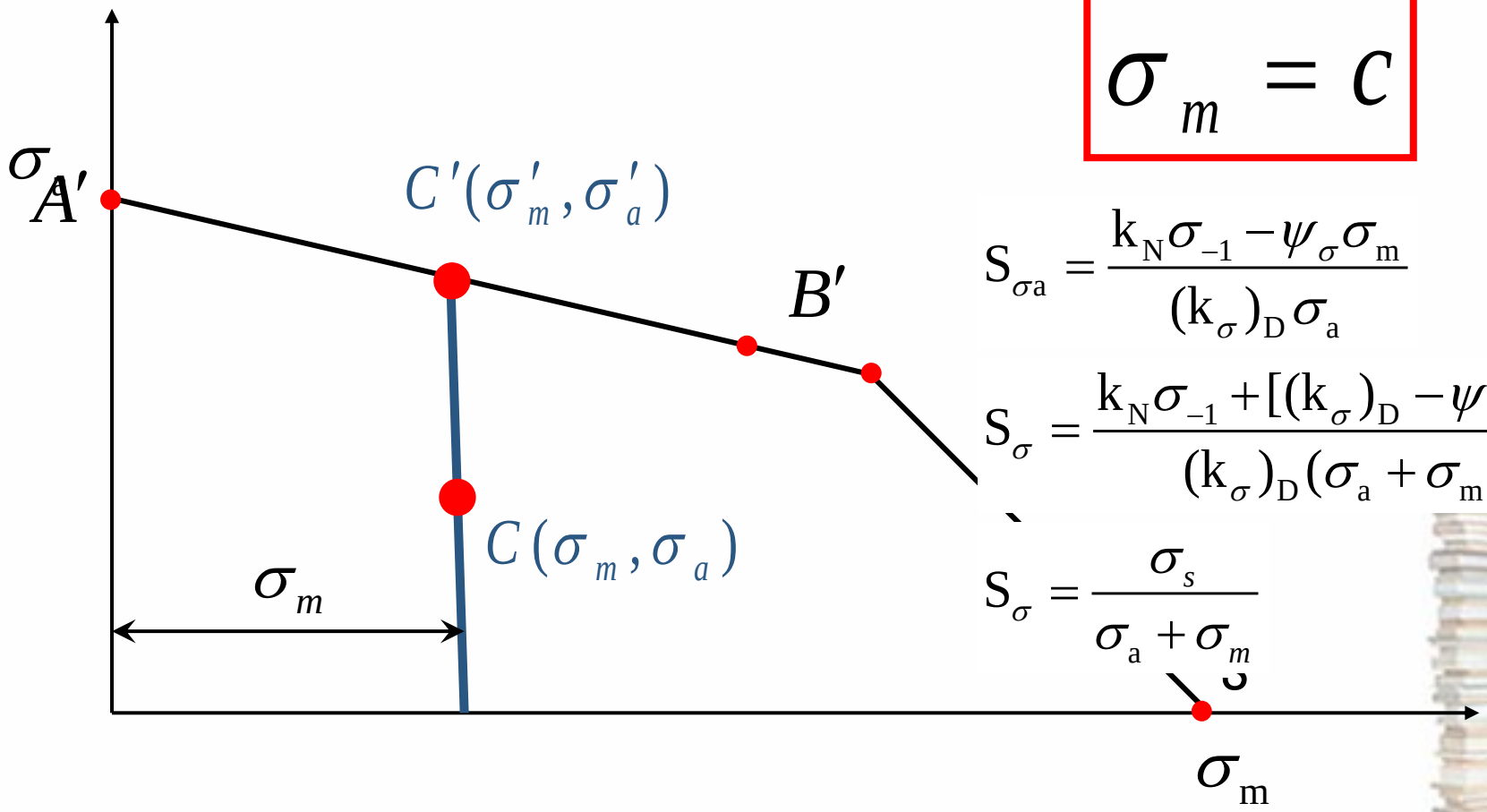
$$r = C$$



这里需要校核
静强度下安全系数

$$S_\sigma = \frac{\sigma_s}{\sigma_a + \sigma_m}$$

3.4 许用疲劳极限应力图



$$\boxed{\sigma_m = C}$$

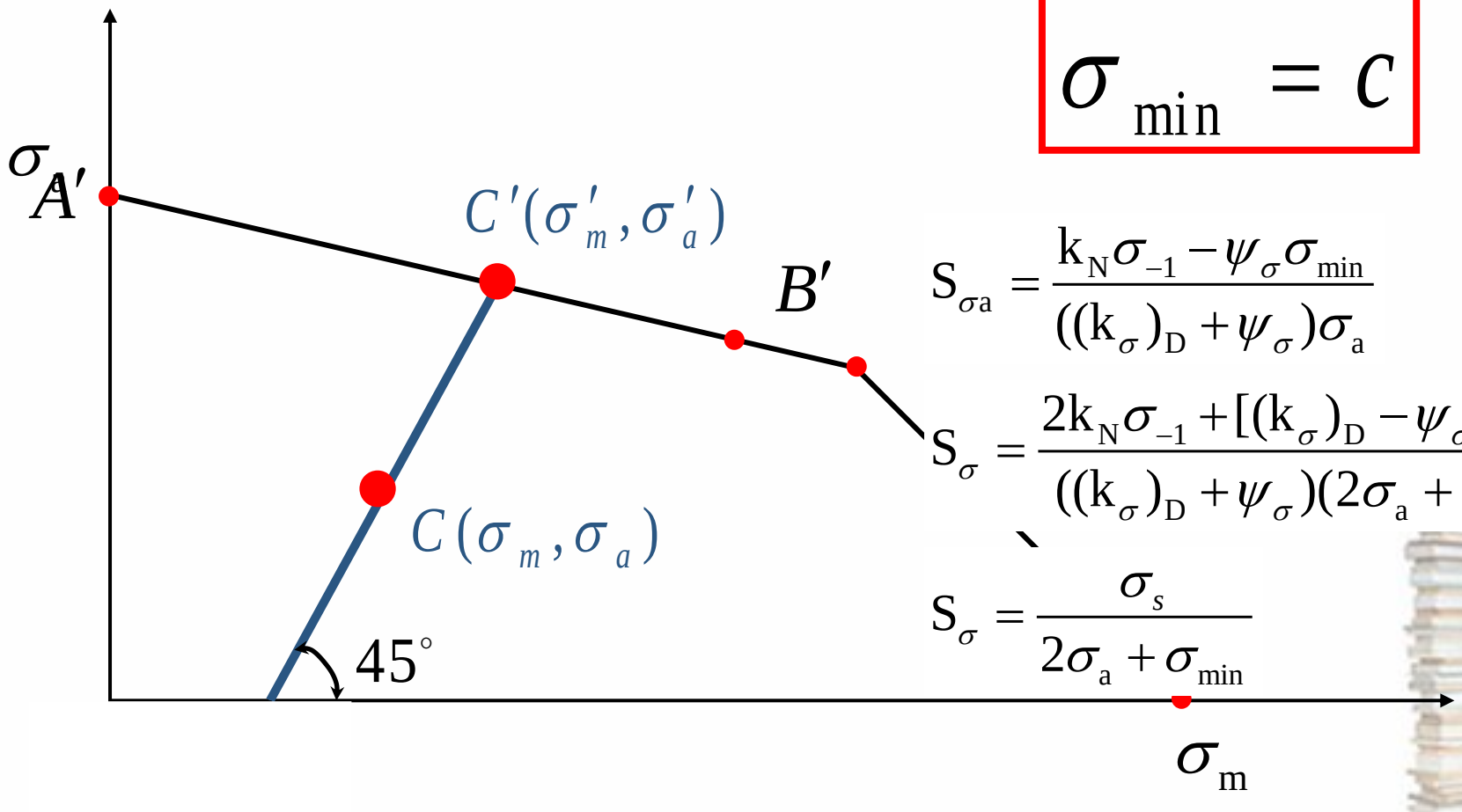
$$S_{\sigma a} = \frac{k_N \sigma_{-1} - \psi_\sigma \sigma_m}{(k_\sigma)_D \sigma_a}$$

$$S_\sigma = \frac{k_N \sigma_{-1} + [(k_\sigma)_D - \psi_\sigma]\sigma_m}{(k_\sigma)_D (\sigma_a + \sigma_m)}$$

$$S_\sigma = \frac{\sigma_s}{\sigma_a + \sigma_m}$$

3.4 许用疲劳极限应力图

$$\sigma_{\min} = C$$



$$S_{\sigma a} = \frac{k_N \sigma_{-1} - \psi_\sigma \sigma_{\min}}{((k_\sigma)_D + \psi_\sigma)\sigma_a}$$

$$S_\sigma = \frac{2k_N \sigma_{-1} + [(k_\sigma)_D - \psi_\sigma]\sigma_{\min}}{((k_\sigma)_D + \psi_\sigma)(2\sigma_a + \sigma_{\min})}$$

$$S_\sigma = \frac{\sigma_s}{2\sigma_a + \sigma_{\min}}$$